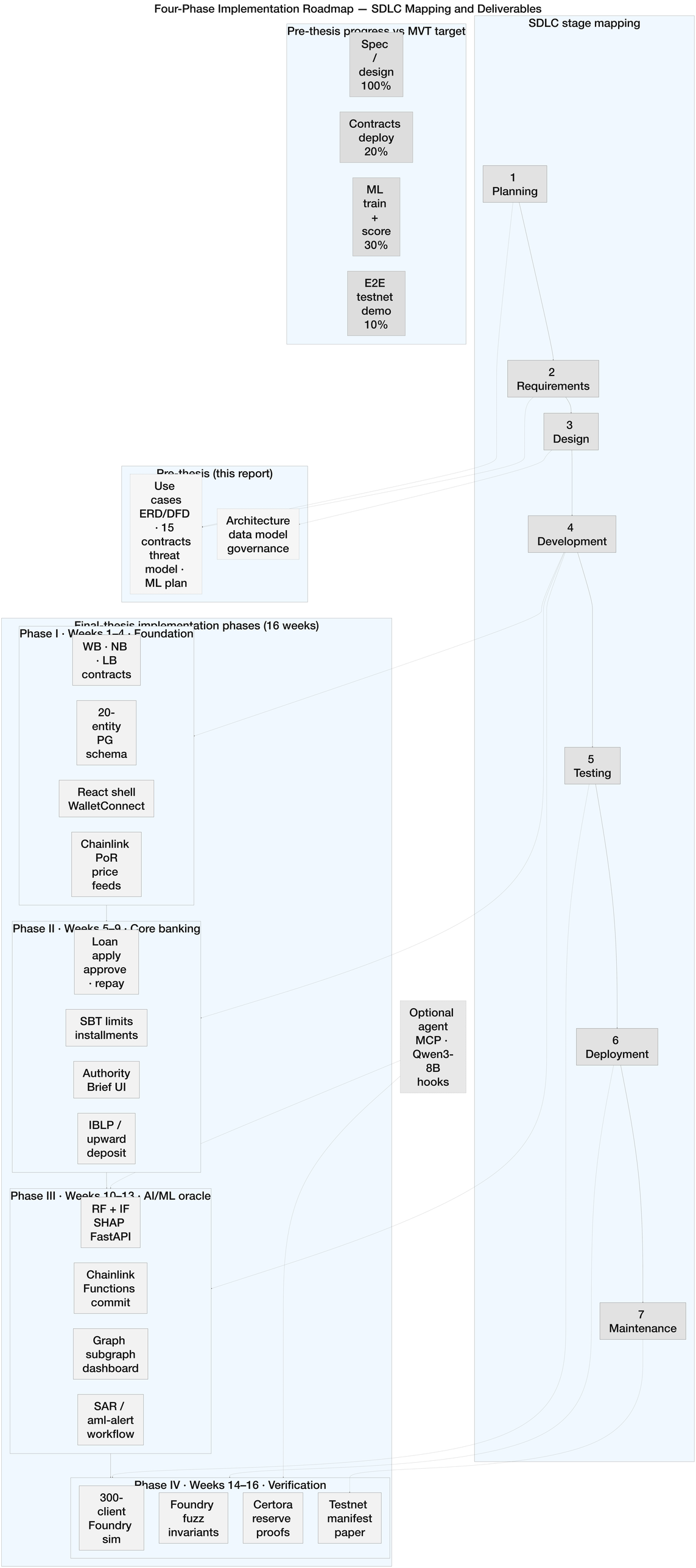
Four-Phase Implementation Roadmap — SDLC Mapping and Deliverables
Pre-thesis progress vs MVT target
Spec / design 100%
Contracts deploy 20%
ML train + score 30%
E2E testnet demo 10%
SDLC stage mapping
1 Planning
2 Requirements
3 Design
4 Development
5 Testing
6 Deployment
7 Maintenance
Pre-thesis (this report)
Use cases ERD/DFD · 15 contracts threat model · ML plan
Architecture data model governance
Final-thesis implementation phases (16 weeks)
Phase I · Weeks 1–4 · Foundation
WB · NB · LB contracts
20-entity PG schema
React shell WalletConnect
Chainlink PoR price feeds
Phase II · Weeks 5–9 · Core banking
Loan apply approve · repay
SBT limits installments
Authority Brief UI
IBLP / upward deposit
Phase III · Weeks 10–13 · AI/ML oracle
RF + IF SHAP FastAPI
Chainlink Functions commit
Graph subgraph dashboard
SAR / aml-alert workflow
Optional agent MCP · Qwen3-8B hooks
Phase IV · Weeks 14–16 · Verification
300-client Foundry sim
Foundry fuzz invariants
Certora reserve proofs
Testnet manifest paper